Scalability & Future Plan

Date	2nd July 2026
Team ID	xxxxxx
Project Name	CreditCardX: Smart Credit Card Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static CSV dataset and does not process real-time or continuous application data.	Predictions may not reflect the latest user behavior and market trends.	high
2	No user authentication or applicant history tracking.	Cannot maintain user profiles or previous prediction/approval history.	medium
3	Model is trained on limited features and may not generalize well.	May lead to lower accuracy for diverse applicant profiles.	medium
4	Deployed on a single local server with limited resources.	Cannot handle high traffic or concurrent users effectively.	high
5	Basic input validation and no robust security mechanisms.	Risk of invalid inputs, data breaches, and unauthorized access.	high

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local Flask server.	Deploy on Render cloud platform and use Gunicorn WSGI server to handle more concurrent users efficiently.
2	Data Storage	Uses a local CSV dataset and no database for storing user data or predictions.	Integrate a relational database such as MySQL or PostgreSQL to store user information, application history, and prediction results.
3	Performance	Prediction model runs on a single machine without caching.	Optimize the ML model, implement caching (e.g., Redis), and use asynchronous processing for better response time and scalability.
4	Security	Basic input validation with no user authentication.	Add user authentication, HTTPS, input sanitization, and role-based access control to enhance application security.

2	Data Storage	Uses a local CSV dataset and a saved ML model (model.pkl).	Integrate a database such as MySQL or PostgreSQL to store user information, prediction history, and future datasets.
3	Performance	Predictions are generated using a single local machine.	Optimize the ML model, implement caching, and upgrade server resources to improve response time and support higher traffic.
4	Security	Basic input validation with no user authentication.	Add user authentication, HTTPS, secure input validation, and role-based access control to improve application security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Requirement analysis, dataset collection, and project planning	Week 1 (Day 1–2)	Establishes a strong foundation for project development.
Phase 2	Data preprocessing, machine learning model training, and model selection	Week 1 (Day 3–4)	Produces an accurate credit card approval prediction model.
Phase 2	Frontend and backend development using HTML, CSS, Python, and Flask (deployed on Render with Gunicorn)	Week 1 (Day 5–6)	Provides a functional web application for credit card approval prediction.
Phase 3	Model integration, testing, bug fixing, documentation, and final demonstration	Week 1 (Day 7)	Delivers a fully functional and tested CreditCardX application ready for presentation.